

Beta particle transport and its impact on betavoltaic battery modeling

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HIGHLIGHTS

- The impact on beta particle penetration depth and energy deposition for betavoltaic battery design are studied using MCNP.
- Different beta particle transport approaches are examined for betavoltaic battery by junction depth analysis.
- An isotropic source needs to be considered in the beta particle transport model.
- The inclusion of the self-absorption effect improves the betavoltaic battery model significantly.

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ABSTRACT

Simulation of beta particle transport from a Ni-63 radioisotope in silicon using the Monte Carlo N-Particle (MCNP) transport code for monoenergetic beta particle average energy, monoenergetic beta particle maximum energy, and the more precise full beta energy spectrum of Ni-63 were demonstrated. The beta particle penetration depth and the shape of the energy deposition varied significantly for different transport approaches. A penetration depth of $2.25 \pm 0.25 \mu\text{m}$ with a peak in energy deposition was found when using a monoenergetic beta particle average energy and a depth of $14.25 \pm 0.25 \mu\text{m}$ with an exponential decrease in energy deposition was found when using a full beta energy spectrum and a 0° angular variation. For a 90° angular variation, i.e. an isotropic source, the penetration depth was decreased to $12.75 \pm 0.25 \mu\text{m}$ and the backscattering coefficient increased to 0.46 with 30.55% of the beta energy escaping when using a full beta energy spectrum. Similarly, for a 0° angular variation and an isotropic source, an overprediction in the short circuit current and open circuit voltage solved by a simplified drift-diffusion model was observed when compared to experimental results from the literature. A good agreement in the results was found when self-absorption and isotope dilution in the source was considered. The self-absorption effect was 15% for a Ni-63 source with an activity of 0.25 mCi. This effect increased to about 28.5% for a higher source activity of 1 mCi due to an increase in thickness of the Ni-63 source. Source thicknesses of approximately $0.1 \mu\text{m}$ and $0.4 \mu\text{m}$ for these Ni-63 activities predicted about 15% and 28.5% self-absorption in the source, respectively, using MCNP simulations with an isotropic source. The modeling assumptions with different beta particle energy inputs, junction depth of the semiconductor, backscattering of beta particles, an isotropic beta source, and self-absorption of the radioisotope have significant impacts in betavoltaic battery design.

1. Introduction

Betavoltaic batteries harvest energy from beta emitting radioisotope sources using semiconductors. The kinetic energy of the beta particles from the radioisotope is converted into electrical energy in the semiconductor. Betavoltaic batteries have some unique features such as a long service life (several years to several decades) and small size (Kwon and Robertson, 2009; Lal and Blanchard, 2004). They can be miniaturized to micron sizes due to their high energy density. These advantages of betavoltaic batteries over other types of batteries make

them an attractive power source for fulfilling the requirements of future generation electronic devices (Wu et al., 2011; Prelas et al., 2014).

This work examines and contrasts the strengths and weaknesses of the main modeling philosophies applied to betavoltaic batteries. The analysis of the energy deposition of beta particles in semiconductors, the estimation of their penetration depth, and the profile of electron-hole pair (EHP) generation in semiconductors are of utmost importance in betavoltaic battery design. These analyses are important in order to get a better estimate of EHP production, which determines the semiconductor design parameters such as top layer thickness or junction

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depth, doping concentrations, and thickness. The design of the battery varies depending on how the beta particle energy is chosen such as average beta particle energy or the full beta energy spectrum, beta particles angular emission such as isotropic or non-isotropic emission, and the consideration of self-absorption effects (Prelas et al., 2016). Many researchers have used different methods such as the Katz-Penfold range equation, the Everhart and Hoff range equation, the Kanaya-Okayama model, and Monte Carlo methods such as Monte Carlo N-Particle (MCNP) transport code for determining energy deposition and penetration depth (Alam and Pierson, 2016; Ulmen et al., 2009; Thomas et al., 2016; Alam et al., 2015). Most of the available betavoltaic designs have only considered the average beta particle energy. The Monte Carlo method using MCNP software (Goorley, 2012) was used in this work because of its comprehensive atomic physics model which obtains better estimates of energy deposition and penetration depth. Monoenergetic beta particle average energy, monoenergetic beta particle maximum energy, and a full beta energy spectrum of Ni-63 were considered as different types of source inputs in the MCNP model to analyze the energy deposition and penetration depth in silicon (Si). Some recent works showed the qualitative difference in energy deposition for different beta input energies and discussed the importance of modeling full beta energy spectrum (Zuo et al., 2013; Zaitsev et al., 2014; Gui et al., 2016; Polikarpov and Yakimov, 2015; Yakimov, 2016). In this work, it will be shown that different input energy profiles could have a significant impact on betavoltaic battery design based on the junction depth analysis. The penetration depths obtained from the MCNP model and the Katz-Penfold range equation for monoenergetic beta particle average energy and monoenergetic beta particle maximum energy of Ni-63 in Si were compared. The MCNP model was further analyzed for different angular distributions of the beta particle's initial direction for a full beta energy spectrum. In reality, beta particles are emitted isotropically from a radioisotope source. It will be shown that the variation in angular distribution of the beta particles is an important factor for betavoltaic battery design. The analysis of a backscattering effect due to various angular emissions of beta particles has significant impact in the design. A simplified drift-diffusion semiconductor model was then used to calculate the short circuit current, the open circuit voltage, and the leakage current for the betavoltaic batteries. It was shown that the junction depth is a very important design parameter in determining the short circuit current, open circuit voltage, and leakage current. However, the optimum junction depth depends on the beta particle energy input and its angular distribution. The simulated betavoltaic battery results were then compared to some results from the literature for actual betavoltaic batteries. The results were explained by the self-absorption effect of the beta particles in the radioisotope source. The self-absorption has normally been overlooked in the theoretical modeling of betavoltaic battery design. In reality, the efficiency of a betavoltaic battery is limited by the self-absorption effect of the radioactive source. Therefore, consideration of the self-absorption effect will provide a better estimate of betavoltaic battery outputs such as the short circuit current, open circuit voltage, and leakage current. It was noted that Zuo et al. also modeled Ni-63 with silicon and validated their results. However, the detailed studies and analyses of beta particles angular emission, backscattering effect, penetration depths, self-absorption effect, and the importance of optimum junction depth are presented in this work. Furthermore, the self-absorption effect of the source was analyzed using MCNP and compared with the results obtained by the chord method (Gui et al., 2016).

The organization of the paper includes a brief summary of the MCNP method and a brief discussion of the beta particle energy spectrum used in this work. It also includes the statistical reliability analysis of the MCNP outputs for convergence. Finally, the important effects of the penetration depth, EHP estimates, angular variation, junction depth, and self-absorption on the short circuit current, open circuit voltage, and leakage current are presented and discussed and then conclusions are drawn.

2. Method

The electron-photon-positron transport feature of MCNP6 (version Beta 3) was used in this work. MCNP uses a detailed physics model of electron-photon-positron transport. The physical events for electron transport are elastic and inelastic collisions of the incoming electrons (beta particles) emitted from a radioisotope source with the atomic electrons and nuclei in the material. An electron mainly loses its energy by inelastic collisions with the atomic electrons and changes direction by elastic collisions with the electrostatic field of the atomic nuclei (Evans, 1955; Zerby and Keller, 1967). The inelastic collisions with orbital electrons dominate at lower kinetic energies such as less than 10 MeV. In this case, the energy is lost due to atomic excitation and ionization, which in turn creates soft x-rays and knock-on electrons. When the electron energy is higher than a few MeV, the inelastic collisions with both atomic nuclei and electrons become dominant, which causes sudden speed changes in the incoming electrons. In this case, the energy is lost due to the bremsstrahlung effect, which in turn creates photons. A photon loses its energy and changes direction by physical events such as photoelectric effect, coherent and incoherent scattering, and pair production (X-5 Monte Carlo Team, 2003). A positron is created during the pair production event. A positron experiences similar scattering events like electrons and annihilates with electrons. MCNP considers all the physical events for each particle by employing different types of cross sections. In this work, Nickel-63 is the beta emitter which has a maximum beta particle energy of 66.7 keV. Thus, the photon energy loss compared to the atomic excitation and ionization, which is the radiation yield, is on the order of 10^{-4} . MCNP uses the continuous slowing down approximation (CSDA) method to estimate the energy loss by collisional stopping power. It also uses Goudsmit-Saunderson's theory for the multiple scattering method to estimate the angular deflections of electrons using the numerically tabulated Riley cross section for electrons under 256 keV. The Moller cross-section is used in MCNP to calculate the generation of knock-on electrons.

Electrons are tracked down to 20 eV in this work. MCNP uses two different algorithms such as the condensed history and the single event method depending on the energy of a particle (Pelowitz, 2013; Hughes, 2012, 2013). The condensed history algorithm is applicable for the energy range of 1 keV to 100 GeV (Pelowitz, 2013), where it condenses multiple electron interactions in each logarithmic energy step size. It is computationally expensive for the electron transport to track all the interactions as it goes through a large number of collisions due to long range coulomb interactions for small energy loss (Hughes, 1996). Thus, the net effect of energy loss and angular deflection are estimated in the condensed history algorithm within a small energy step size. On the other hand, MCNP uses the single event method for particles with energy below 1 keV to 10 eV (Hughes, 2013). In the single event method, MCNP calculates the path lengths for all four physical events such as elastic scattering, atomic excitation, knock-on, and bremsstrahlung, then, takes the shortest path length of the associated physical events. It estimates the energy loss and angular deflection from the cross sections of the selected physical events. The average behavior of the physical system such as the energy deposition was inferred by MCNP from the statistics of individual particles.

3. Precise beta spectrum vs generalized beta spectrum for MCNP input

The amount of energy deposited and the depth of penetration of the beta particles emitted by the radioisotope source depends on their initial kinetic energy. Examples of radioisotopes that undergo radioactive decay by emitting beta particles are Ni-63, Sr-90, S-35, and P-32 (Alam and Pierson, 2016). Each beta particle emitted in decay has a continuous energy spectrum between zero and a maximum value. This maximum beta particle energy varies for different radioisotopes. The result is that each radioisotope will have a continuous energy

probability distribution, which is known as the beta energy spectrum. The beta energy spectrum is derived from Fermi's golden rule based on the weak force interaction in the atomic nucleus (Krane, 1987). The transition probability rate of beta particles, λ , is related to the final density of states by Eqs. (1)–(3),

$$\lambda = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_f), \quad (1)$$

$$V_{fi} = \int \Psi_f^* V \Psi_i d\tau, \quad (2)$$

$$\rho(E_f) = \frac{dn}{dE_f}, \quad (3)$$

where, V_{fi} is the matrix element, which is the integral of all the interactions V between initial and final states, Ψ is the wave function of the initial and final atomic states, and density of the final states, $\rho(E_f)$, represents the number of accessible final states dn within the energy dE_f . The beta energy spectrum is then derived from the allowed approximation of the electron wave function in Fermi's golden rule, where the matrix element has no influence on the shape of the beta energy spectrum, and is given by Eq. (4),

$$N(T_e) \propto (T_e^2 + 2T_e m_e c^2)^{1/2} (Q - T_e)^2 (T_e + m_e c^2), \quad (4)$$

where, T_e is the electron kinetic energy, m_e is the electron mass, Q is the decay energy, and c is the speed of light. This spectrum vanishes at $T_e = 0$ and $T_e = Q$, which is a generalized spectrum with similar shape for all beta emitting radioisotopes based on their maximum beta energy as shown in Fig. 1. This beta energy spectrum shows a peak around the average energy, which is typically one-third of its maximum energy. However, the inclusion of the Fermi function that takes into account coulomb interaction of the beta particles with the atomic nucleus can change this typical generalized spectrum shape. In the case of Ni-63, it exhibits a different type of beta energy spectrum, where the peak of the probability distribution of the beta particle energy is observed at low beta particle energy instead of at the average beta particle energy (Cross et al., 1983; ICRP, 2008). The probability distribution then decreases with increasing beta particle energy and becomes a minimum at the maximum energy as shown in Fig. 1. This is therefore a more accurate representation of the beta energy spectrum for Ni-63 (Alam et al., 2015).

It is important to note that many previous simulations (Guo and Lal, 2003; Zhao et al., 2011) of betavoltaic batteries used either a single monoenergetic average beta particle energy or the maximum beta particle energy. This significantly affects the resulting energy deposition profile and penetration depth estimates. As a result, the simulated outputs of these models will not be accurate. This phenomenon is also emphasized by some recent works (Zuo et al., 2013; Zaitsev et al., 2014; Gui et al., 2016; Polikarpov and Yakimov, 2015; Yakimov, 2016; Oh et al., 2012). More specifically, the qualitative difference in energy deposition for monoenergetic average beta energy and full beta energy spectrum was highlighted using in-house Monte Carlo simulation and scanning electron microscope (Polikarpov and Yakimov, 2015). Thus, an accurate continuous beta energy spectrum will be used for comparison in all of the simulations presented in this paper.

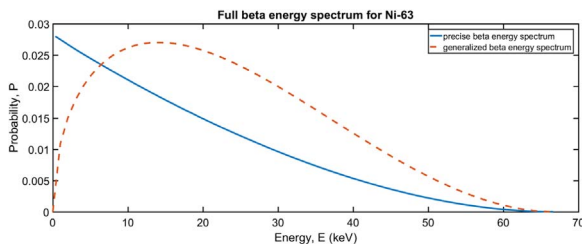


Fig. 1. Different types of continuous beta energy distributions for Ni-63.

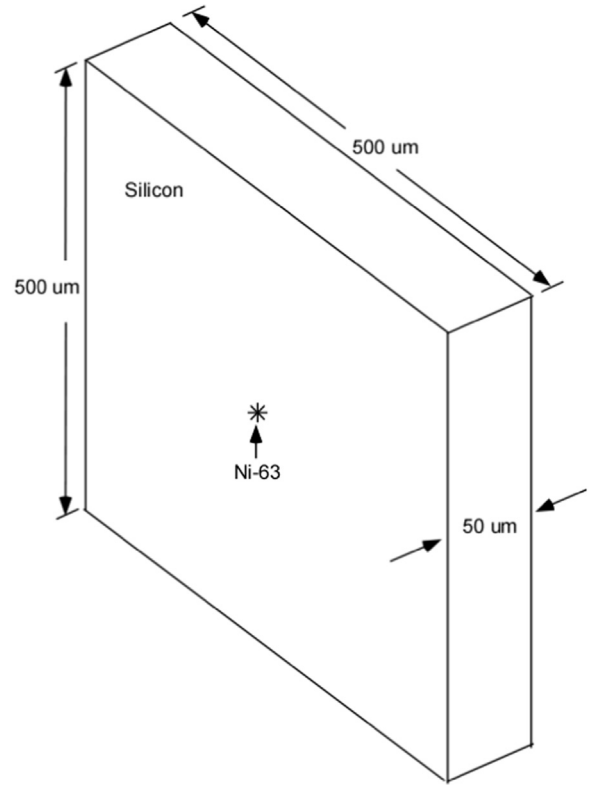


Fig. 2. MCNP simulation setup for a Ni-63 point source with silicon slab geometry.

4. Convergence and reliability of MCNP simulation results

An MCNP simulation using a Ni-63 full beta energy spectrum with silicon was analyzed. The energy deposition F6 tally was scored for a Ni-63 point source placed on the side wall of a slab geometry of silicon with a dimension of $500 \times 500 \times 5 \mu\text{m}$ as shown in Fig. 2. The simulation was run with different particle histories of 10^3 , 10^4 , 10^5 , 10^6 , 10^7 , and 10^8 to study the convergence of the energy deposition tally and the reliability of the results.

For the MCNP simulation, the silicon slab was divided into 100 cells, each $0.5 \mu\text{m}$ thick for a spatial resolution of $\pm 0.25 \mu\text{m}$, in the thickness direction in order to achieve a fine tally resolution. The volumetric energy deposition of each cell was tallied along the thickness direction until the full $50 \mu\text{m}$ thickness was reached. As expected, the energy deposition was very high near the surface and dropped exponentially along the thickness direction. It is noted that as further penetration is achieved into the silicon, the energy deposition eventually becomes less than that needed for electron hole pair formation. This point is reached after about $21 \mu\text{m}$ thickness. No electron hole pair will be generated beyond this maximum penetration depth. However, the design will be optimized for a thickness where 99% of the total energy of the incoming beta particles is deposited. This depth will be less than the maximum possible penetration depth, but does not result in any loss of performance in the battery.

To study the reliability and convergence of the energy deposition tally, the relative errors were plotted against the penetration depth for different particle histories. Fig. 3 shows the relative error of each tally score for all particle histories up to a thickness of $21 \mu\text{m}$. The plot indicates that the energy deposition tally is converging since the relative error decreases with an increasing number of particles. The relative error also increases along the thickness direction, since there are fewer particles remaining at greater depths which reduce the statistical accuracy. However, the relative error is still less than 0.045 (or 4.5%) even at the maximum penetration depth for 10^6 , 10^7 , and 10^8 particle histories. For 10^7 and 10^8 particle histories, the relative errors are

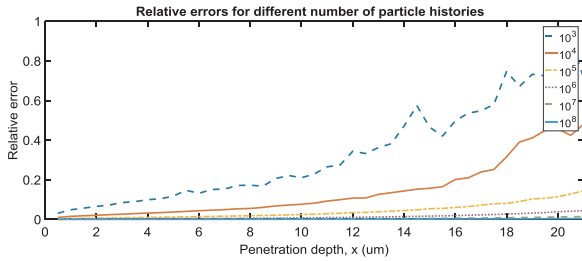


Fig. 3. Relative error vs penetration depth for different particle histories.

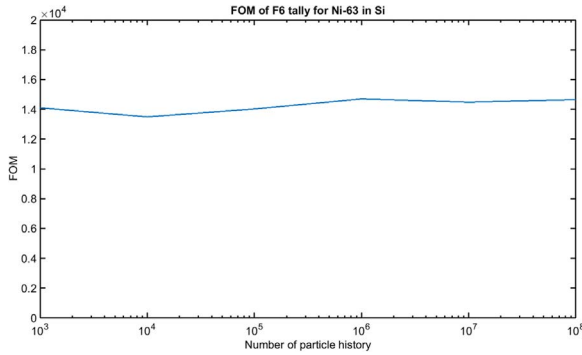


Fig. 4. FOM vs number of particle histories.

0.0145 (or 1.45%) and 0.0046 (or 0.46%), respectively, at the maximum penetration depth.

Fig. 4 shows the variation in the Figure of Merit (FOM) with different particle histories. FOM is expressed by $1/R^2T$, where R is the relative error and T is the computation time. It is a dimensionless number, where a higher number indicates a more efficient simulation. The relative error can be reduced with the expense of higher computation time by increasing the number of particle histories. In Fig. 4, FOM is almost constant for a large variation in particle histories. This implies that the energy deposition (F6 tally) results are stable.

Both Figs. 3 and 4 indicate that the energy deposition tally is converging. From this analysis of the relative error, a 10^7 particle history is chosen to provide reliable statistics for the energy deposition tally. A semi-log plot of energy deposition with uncertainties for 10^7 particle histories is shown in Fig. 5. As the uncertainties are very low, the vertical error bars with maximum and minimum are shown as horizontal lines in Fig. 5.

Fig. 5 shows that the energy deposition decreases exponentially with thickness and uncertainty of the results is insignificant along the penetration depth of 21 μm . Therefore, the energy deposition using 10^7 number of particle histories is reliable for this analysis and will be used in the rest of the simulations.

5. Analysis of beta particle transport results using MCNP

To compare the effects of using different beta particle energies, a similar geometry and source setup like that in Fig. 2 was used to

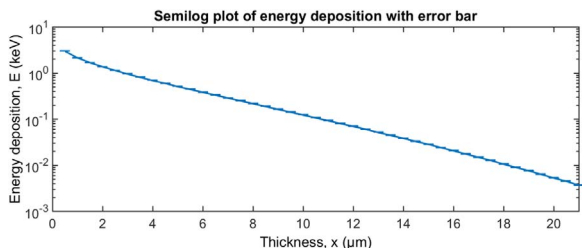


Fig. 5. Energy deposition along the thickness direction for Ni-63 in Si.

determine the penetration depth in silicon for three different Ni-63 energy cases. These cases used (1) the monoenergetic average beta particle energy for Ni-63, (2) the monoenergetic maximum beta particle energy for Ni-63, and (3) the full beta energy spectrum (precise) for Ni-63 shown in Fig. 1. Since 10^7 particle histories showed convergence and a low relative error for this slab configuration in the previous section, this number of particle histories was used in this analysis for the different beta particle energy cases listed above. The penetration depth of electrons in a material will vary with the different input energies. The penetration depth is the maximum thickness required for a material to stop the majority of the beta particles. It can be determined by observing the depth at which the MCNP energy deposition values become negligible. From a betavoltaic battery perspective, any energy deposition amount that is less than the energy required to create one electron hole pair can be considered negligible. However, from an engineering design perspective, the energy deposition beyond 99% of the total energy deposition can be neglected. The penetration depths for Ni-63 in silicon are determined based on these two criteria. The penetration depths from the MCNP energy deposition simulation results are compared in Table 1 for the three different input energy cases described above. For the case where the Ni-63 full beta energy spectrum is used, the Table also provides results based on the angular dependency of the initial direction of the emitted beta particles. The beta particles conical distribution or angular direction, θ , emitted from a radioisotope source is shown in Fig. 6. An actual radioisotope source emits the beta particles in an isotropic direction. In other words, the beta particles are not emitted only in the straight-line x-axis direction as the first three rows of results imply. To simulate the effect of isotropic emission of beta particles from the Ni-63 source, the third case for the full energy spectrum was modified to show three different isotropic spreading cones: 30°, 60°, and 90°. The 90° cone should provide a more realistic estimate of how the beta particles are emitted toward the semiconductor material used in betavoltaic batteries. For each of these cones, the total source strength, in this case the same number of particle histories, was emitted over the spherical angle indicated. As expected, this angular emission reduced the penetration depth somewhat. For the two cases involving monoenergetic beta particle energies, a comparison was made by estimating the penetration depth using the empirical Katz-Penfold equation for low Z materials (Katz and Penfold, 1952; Turner, 2008), which is given in Eq. (5),

$$R = 0.412T^{1.27-0.0954\ln T}, (0.01 \leq T \leq 2.5\text{MeV}) \quad (5)$$

where R is the range density in g/cm^2 and T is the electron kinetic energy in MeV. R is then divided by the density of silicon to get the penetration depth. Note that this equation assumes the beta particles are emitted in the straight-line x-axis direction. The Katz-Penfold predictions are shown in the far right column of Table 1. These values are close to those predicted by MCNP for the case where 99% of the total beta particle energy has been deposited. However, as expected, the Katz-Penfold estimated values are quite different than those predicted by MCNP when the full beta energy spectrum is used.

The determination of penetration depth using two different criteria indicates that a considerable amount of additional thickness, which is about 6 μm for the full beta energy spectrum, is required to absorb a negligible amount of energy. The results show a slight dependency of penetration depth based on angular variation of the initial direction of flight of the beta particles. The penetration depth decreases with higher initial angular direction of flight of the beta particles. In reality, since beta particles emit a continuous beta energy spectrum in a random direction from a radioisotope, the predicted penetration depth of $12.75 \pm 0.25 \mu\text{m}$ for Ni-63 beta particles in silicon provides a better estimate.

In addition, the backscattering coefficient, which is the percentage of beta particles scattered back from the surface, depends on the angular variation of the initial direction of impinging beta particles. A point source and a silicon surface that had a much larger cross-sectional

Table 1
Variation of penetration depth for different energy inputs.

Input type	Penetration depth up to 1 EHP energy (μm)	Penetration depth up to 99% energy deposited (μm)	Penetration depth by Katz-Penfold (μm)
Monoenergetic beta average energy (17.4 keV) 0°	3.75 ± 0.25	2.25 ± 0.25	2.16
Monoenergetic beta maximum energy (66.9 keV) 0°	37.25 ± 0.25	29.25 ± 0.25	28.49
Full beta energy spectrum (0–66.9 keV) 0°	20.75 ± 0.25	14.25 ± 0.25	
Beta energy spectrum 30° cone	20.25 ± 0.25	14.25 ± 0.25	
Beta energy spectrum 60° cone	19.25 ± 0.25	13.25 ± 0.25	
Beta energy spectrum 90° cone	17.75 ± 0.25	12.75 ± 0.25	

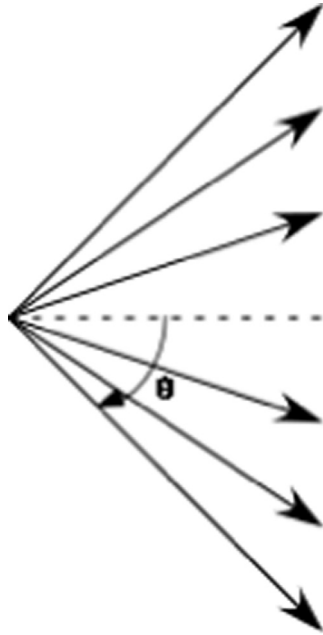


Fig. 6. Conical distribution of beta particles in different angular directions.

area of $500 \times 500 \mu\text{m}^2$ compared to the MCNP penetration depth was used to calculate the backscattering coefficient in terms of the particles weight lost from the impinging surface.

Fig. 7 shows that the backscattering coefficient increases with increasing angle of the initial direction of flight of a single beta particle. It also shows the percentage of the beta particle energy lost due to the backscattering effect of the surface. For 0° to 90° , the backscattering coefficient varies from 0.16 ± 0.001 to 0.426 ± 0.002 and the percentage of energy lost varies from $8.92 \pm 0.06\%$ to $30.42 \pm 0.1\%$ respectively. It indicates that more beta particles with lower energies are lost due to the backscattering effect. It means that even though beta particles that are

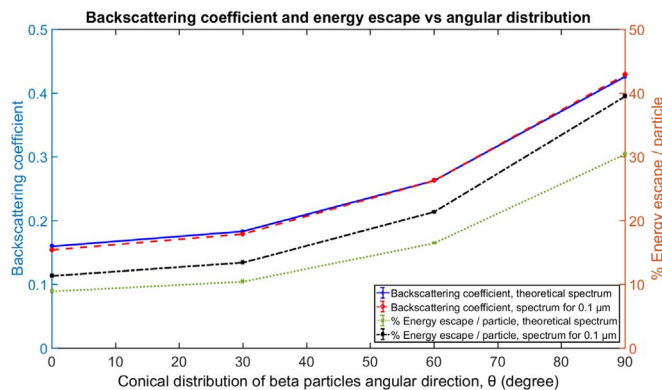


Fig. 7. Backscattering effect with different initial beta particle direction of flight.

emitted at a large angle contribute most to the efficiency by generating more EHPs in the depletion region (Theirattanakul and Prelas, 2017), they will also have a large electron backscattering effect which reduces efficiency. Furthermore, it was noted that the theoretical beta energy spectrum changes with the source thickness (Gui et al., 2016; Li et al., 2012). Fig. 7 also shows the backscattering coefficient and energy lost for the modified spectrum of a $0.1 \mu\text{m}$ thick Ni-63 source. The backscattering coefficient was almost the same for the modified spectrum. However, the energy escape increased due to higher source thickness. Thus, the angular distribution of beta particles needs to be considered in the model in order to take into account the energy lost due to the backscattering effect for a better estimate of energy deposition. This will be discussed further in the next section.

The plots of energy deposition in silicon vary significantly for the monoenergetic average beta particle energy, the monoenergetic maximum beta particle energy, and the full beta particle energy spectrum cases for Ni-63 as seen in Fig. 8.

A peak in the energy deposition plot occurs in Fig. 8 for both the monoenergetic average and maximum beta particle energies. On the other hand, the energy deposition plot along the thickness direction is almost exponential for the full beta particle energy spectrum (Zuo et al., 2013). Therefore, in this latter case, most of the energy is deposited near the surface. Furthermore, the energy deposition near the surface increases with a higher angular beta particle direction and their penetration depth decreases. Increased angular direction makes the exponential curve for the energy deposition plot steeper.

Thus, an MCNP analysis of beta particle transport provides a better estimate for energy deposition in the semiconductor due to its detailed physics model. This variation in the angular distribution of beta particles with a full beta energy spectrum is incorporated to provide a more accurate betavoltaic battery model. The energy deposition estimate determines the semiconductor device design parameters for the betavoltaic battery, which will be discussed in the following section.

6. Semiconductor device model for betavoltaic battery

MCNP only provides the beta particle energy deposition and does

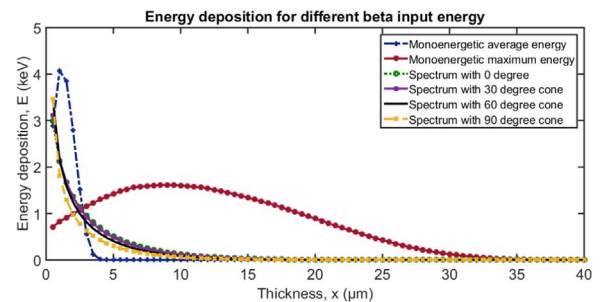


Fig. 8. Energy deposition for monoenergetic average beta particle energy, monoenergetic maximum beta particle energy, and full beta energy spectrum with different initial beta particle direction of flight for Ni-63 in Si.

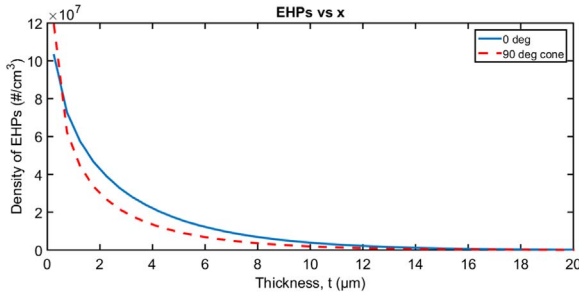


Fig. 9. The generation of EHPs in Si for full beta energy spectrum of Ni-63 with 0° and 90° cone of initial direction.

not provide an estimate of EHP generation in semiconductors. The electron hole pair ionization energy of 3.64 eV for silicon was used in a linear relationship to calculate the number of EHPs generated from the energy deposition as shown in Fig. 9. As a result, the plot for EHP generation in a semiconductor has the same shape as the energy deposition plot from MCNP. After calculating the generation of EHPs from the MCNP energy deposition results, a semiconductor model is then required to estimate the collection of EHPs at the electrodes for the betavoltaic battery output. A closed form solution of the semiconductor drift-diffusion model with Boltzmann approximation was used to calculate the short circuit current, the open circuit voltage, and the leakage current of the betavoltaic battery. The drift current due to the electric field was considered only in the depletion region, whereas the diffusion equations were solved in p-type and n-type regions for the minority carriers. Finally, the results were compared to experimental results with a betavoltaic battery using Ni-63 and a silicon transducer (Guo and Lal, 2003).

Fig. 10 is a sketch of a p-n junction semiconductor used in this betavoltaic battery model. Similar to the experimental battery design, the Ni-63 radioisotope was placed on the outer surface of the p-side of the p-n junction silicon semiconductor. The EHPs that are generated in the semiconductor from the beta particle energy deposition are separated by the built-in electric field at the p-n junction. An exponential approximation of the above EHP generation plot was used in Eqs. (6) and (7) represented by the term $A_0 a e^{-bx}$ or $A e^{-bx}$, where $A = A_0 a$.

$$\frac{d^2 \Delta n}{dx^2} - \frac{\Delta n}{L_n^2} = -\frac{A_0 a e^{-bx}}{D_n} \quad (6)$$

$$\frac{d^2 \Delta p}{dx^2} - \frac{\Delta p}{L_p^2} = -\frac{A_0 a e^{-bx}}{D_p} \quad (7)$$

The diffusion Eqs. (6) and (7) for the minority carriers were solved in the p-type and n-type regions of the semiconductor with two boundary conditions.

$$D_n \frac{dn}{dx} = S_n \Delta n \quad (8)$$

$$\Delta n(x_p) = 0 \quad (9)$$

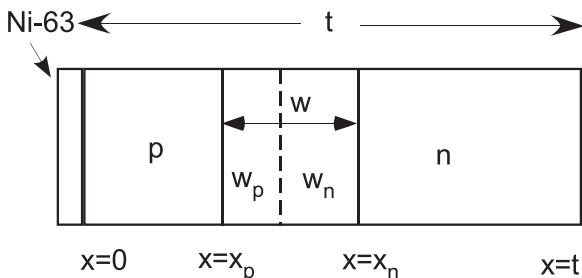


Fig. 10. Drift-diffusion semiconductor model for betavoltaic batteries.

$$-D_p \frac{dp}{dx} = S_p \Delta p \quad (10)$$

$$\Delta p(x_n) = 0 \quad (11)$$

Eqs. (8) and (9) are the boundary conditions for the p-type region, whereas Eqs. (10) and (11) are the boundary conditions for the n-type region. The boundary conditions given in Eqs. (8) and (10) were applied at the surface considering the surface recombination. Similarly, Eqs. (9) and (11) were applied at the edge of the depletion region representing no excess minority carriers at that location. After solving the minority carrier diffusion equations, the short circuit currents in the n-type and p-type regions are given in Eqs. (12) and (13) respectively. The depletion region current in Eq. (14) is the integral of the EHPs in the depletion region.

$$J_n = \left\{ \frac{q A L_n}{b^2 L_n^2 - 1} \right\} \left[\frac{\left(\frac{S_n L_n}{D_n} + b L_n \right) - e^{-bx_p} \left\{ \frac{S_n L_n}{D_n} \cosh\left(\frac{x_p}{L_n}\right) + \sinh\left(\frac{x_p}{L_n}\right) \right\}}{\frac{S_n L_n}{D_n} \sinh\left(\frac{x_p}{L_n}\right) + \cosh\left(\frac{x_p}{L_n}\right)} - b L_n e^{-bx_p} \right] \quad (12)$$

$$J_p = \left\{ \frac{q A L_p}{b^2 L_p^2 - 1} \right\} \left[\frac{\left(\frac{S_p L_p}{D_p} - b L_p \right) e^{-bt} - e^{-bx_n} \left\{ \frac{S_p L_p}{D_p} \cosh\left(\frac{t-x_n}{L_p}\right) + \sinh\left(\frac{t-x_n}{L_p}\right) \right\}}{\frac{S_p L_p}{D_p} \sinh\left(\frac{t-x_n}{L_p}\right) + \cosh\left(\frac{t-x_n}{L_p}\right)} + b L_p e^{-bx_n} \right] \quad (13)$$

$$J_{dep} = \frac{qA}{b} e^{-bx_p} [1 - e^{-bw}] \quad (14)$$

The short circuit current of the battery is the sum of the short circuit currents in the p-type, n-type, and depletion regions. The short circuit current is directly proportional to the activity of the radioisotope. The open circuit voltage of the battery and the leakage current of the semiconductor are given in Eqs. (15) and (16) respectively.

$$V_{oc} = \frac{kT}{q} \ln \left(\frac{J_{sc}}{J_0} + 1 \right) \quad (15)$$

$$J_0 = \frac{q D_n n_{p0}}{L_n} \left[\frac{\frac{S_n L_n}{D_n} \cosh\left(\frac{x_p}{L_n}\right) + \sinh\left(\frac{x_p}{L_n}\right)}{\frac{S_n L_n}{D_n} \sinh\left(\frac{x_p}{L_n}\right) + \cosh\left(\frac{x_p}{L_n}\right)} \right] + \frac{q D_p p_{n0}}{L_p} \left[\frac{\frac{S_p L_p}{D_p} \cosh\left(\frac{t-x_n}{L_p}\right) + \sinh\left(\frac{t-x_n}{L_p}\right)}{\frac{S_p L_p}{D_p} \sinh\left(\frac{t-x_n}{L_p}\right) + \cosh\left(\frac{t-x_n}{L_p}\right)} \right] \quad (16)$$

The open circuit voltage is dependent on the ratio of the short circuit current to the leakage current. In Eqs. (6)–(16), n is the minority carrier concentration in the p-type region, p is the minority carrier concentration in the n-type region, L is the minority carrier diffusion length, D is the diffusion coefficient, A_0 is half of the total source activity assuming that only half of the beta particle emissions are in the direction of the semiconductor surface since the source is isotropic, S is the surface recombination, and a and b are the coefficients for the EHP generation plot exponential curve fitting. The values of the parameters used in this work for silicon at a temperature of 300 K are listed in Table 2.

The junction depth in the silicon was varied to estimate the short circuit current densities, open circuit voltage, and the leakage current densities for the Ni-63 full beta particle energy spectrum with a 0° and a 90° cone of initial beta particle direction. The resulting variation in

Table 2
Parameters used in the model.

Minority carrier diffusion length (μm)		Diffusion coefficient (cm^2/s)		Activity (mCi)		Surface recombination velocity (cm/s)		Coefficients of the exponential curve fitting (ae^{-bx}) of EHPs	
L_n	L_p	D_p	D_n	Case 1	Case 2	S_p	S_n	a (cm^{-3})	b (cm^{-1})
0.68	49	6.02	6.08	0.25 (0.125 mCi on each side)	1 (0.5 mCi on each side)	10^6	10^6	1.03×10^8 (0°); 1.284×10^8 (90°)	3952 (0°); 7123 (90°)

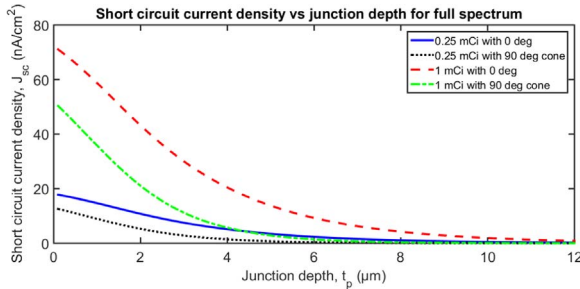


Fig. 11. Variation in short circuit current density with junction depth.

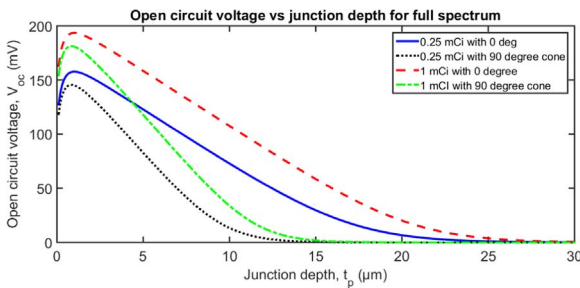


Fig. 12. Variation in open circuit voltage density with junction depth.

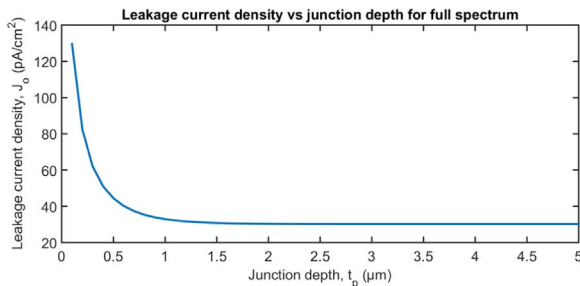


Fig. 13. Variation in leakage current density with junction depth.

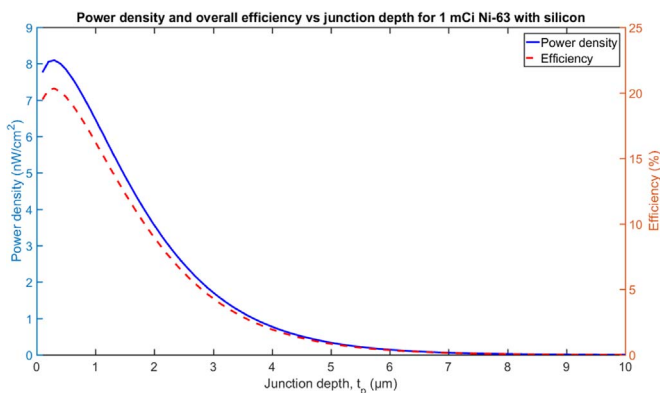


Fig. 14. Variations in power density and efficiency with the junction depth for 90° distribution.

short circuit current density, open circuit voltage density, and the leakage current density is shown in Figs. 11, 12, and 13 respectively. Furthermore, the power density and efficiency analysis with the junction depth is shown in Fig. 14.

The silicon based betavoltaic battery usually provides the largest short circuit current density compared to wide band gap semiconductors. Fig. 11 shows that the short circuit current density decreases with the junction depth and becomes almost zero beyond $12 \mu\text{m}$ junction depth. It is attributed to the fact that the penetration depth for Ni-63 in silicon is about $13\text{--}14 \mu\text{m}$ and the minority carrier diffusion length in p-type region is about $0.68 \mu\text{m}$ due to heavy doping in the p-type region. On the other hand, the minority carrier diffusion length in the n-type region is about $49 \mu\text{m}$, which is much larger than the penetration depth. As a result, the highest short circuit current density is obtained for a junction depth near to the surface as the generations of EHPs are higher due to the maximum energy deposition near to the surface as modeled by the full beta energy spectrum. Furthermore, the short circuit current density is higher near to the surface irrespective of the beta particles angular distribution. However, the magnitude of the short circuit current density for the 90° angular distribution is lower than that for the 0° angular distribution. These differences between the two angular distributions increase with higher activity. Nevertheless, an increased activity increases the short circuit current density regardless of the angular distribution, as the short circuit current is directly proportional to the activity. A more realistic betavoltaic battery model should assume a 90° angular distribution and a junction depth closer to the surface may be required to optimize current output.

The open circuit voltage increases with the junction depth, reaches a peak and then decreases. The peaks in open circuit voltage are obtained at about $1 \mu\text{m}$ and $0.8 \mu\text{m}$ junction depths for 0° and 90° distributions, respectively, regardless of the activities. After the peak, the open circuit voltage decreases almost linearly with the junction depth. The decrease in the ratio of the short circuit current to the leakage current could be the reason for low open circuit voltage with higher junction depth. For 0° and 90° angular distributions, the open circuit voltage becomes almost zero beyond about $25 \mu\text{m}$ and $15 \mu\text{m}$ respectively. With increased activity, the open circuit voltage density curve has the same shape except that the magnitude increases with higher activity.

The leakage current density is about three orders of magnitude lower than the short circuit current density. The leakage current density is very high when the junction depth is near to the surface. However, it decreases significantly once the junction depth reaches $0.5 \mu\text{m}$ from the surface. It does not decrease much further once the junction depth is greater than $2 \mu\text{m}$. The leakage current density is not dependent on the radioisotope and its activity. Therefore, the variation in the angular distribution of beta particles does not have any effect on it.

Fig. 14 shows the power output and the device efficiency with the junction depth for the activity of 1 mCi Ni-63 without considering the self-absorption effect in the source. The device efficiency was defined in Eq. (17) by the ratio of the power output obtained from the device to the maximum theoretical power output.

$$\text{Device efficiency} = \frac{P_{\text{out}}}{P_{\text{theor max}}} \quad (17)$$

Fig. 14 indicates that there is an optimum power output and for a junction depth near to the surface at about $0.3 \mu\text{m}$. Therefore, it shows

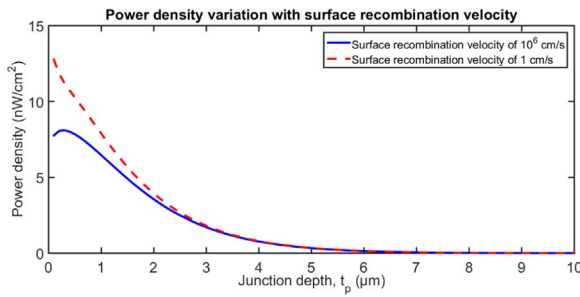


Fig. 15. Surface passivation impact on power output analysis for 90° distribution.

the importance of an optimum junction depth to maximize the power output and efficiency. However, this optimization is dependent on the surface recombination velocity as shown in Fig. 15. For a very well passivated silicon surface, the surface recombination velocity can be reduced significantly (Otto et al., 2012). In this case, a junction depth closer to the surface will provide the maximum power output.

For further analysis, a junction depth of 2 μm was chosen. For this junction depth, a leakage current of 4.8 pA was estimated which agrees well with the theoretical estimates provided by Guo and Lal (2003). However, the experimentally measured leakage current of 24 pA is much higher due to the defects created in the heavily doped p-type glass-source (Guo and Lal, 2003). Guo et al. placed electroplated Ni-63 on a p-type region surface of a silicon p-n junction. They built the silicon p-n junction using doping concentrations of $10^{20}/\text{cm}^3$ and $10^{17}/\text{cm}^3$ in the p-type and n-type regions respectively. They measured the short circuit current and open circuit voltage for 0.25 mCi and 1 mCi of Ni-63. Similar to the experiment, the short circuit current and the open circuit voltage were estimated using the model from above for two different Ni-63 source activities of 0.25 mCi and 1 mCi as shown in Table 3 and Table 4. They were also estimated for two different beta particle angular variations of 0° and 90° in Table 3 and Table 4 respectively. A short circuit current of 1.73 nA and an open circuit voltage of 152 mV were calculated for the case of 0.25 mCi with 0° angular distribution. Similarly, a short circuit current of 6.91 nA and an open circuit voltage of 187.8 mV were calculated for the case of 1 mCi with 0° angular distribution. The short circuit current increased about four times with the use of a factor of four higher activity, as the short circuit current is linearly proportional to the activity. For a junction depth of 2 μm, the short circuit current decreases by a factor of two for the 90° angular distribution. However, the open circuit voltage does not reduce as significantly as the short circuit current density with the angular distribution. Furthermore, the beta particle transport approach such as using the monoenergetic beta particle average energy or the full beta energy spectrum has a significant impact in estimating the junction depth for an optimum battery output. In these cases, the energy deposition peak appears at a different thickness in the semiconductor as shown in Fig. 8. This creates EHPs with a similar profile i.e. the EHP peak is at the same thickness. Ideally, the junction depth is chosen at a thickness where the EHP generation is maximum. Maximum EHP collection occurs in the depletion region around the junction depth. Therefore, a better choice of junction depth that considers the full beta energy spectrum with an isotropic angular distribution will improve the

battery output and provide a higher efficiency.

In Table 3 and Table 4 for all cases, it was observed that the predicted results for the short circuit current and the open circuit voltage were higher than the actually measured short circuit current and open circuit voltage in the experiment. This phenomenon of overprediction of the results (Alam and Pierson, 2016) was also observed for other theoretical models in the literature. It is proposed in this analysis that one of the main reasons for this overprediction of results is due to the self-absorption effect of the beta particles in the source. The self-absorption effect reduces the effective activity of a radioisotope depending on its thickness as some of the beta particle energy deposits in the source before it can exit.

A percentage reduction in the effective activity was applied to the results shown in Table 3 and Table 4 in order to predict the amount of self-absorption needed to match with the short circuit current. The discrepancies in the open circuit voltage are attributed to the p-type glass-source defects explained by Zuo et al. (2013). For 0.25 mCi, 59% and 15% self-absorption effect would explain the observed results for 0° and 90° angular distribution respectively. The self-absorption effect of the radioactive source depends on the surface area and the thickness of the radioisotope. Therefore, when the activity is increased to 1 mCi while keeping the surface area the same, the self-absorption effect further increases as that means the thickness of the source material must increase. As a result, the self-absorption effect increased to about 65.1% and 28.5% for 0° and 90° angular distribution, respectively, for a higher activity of 1 mCi. The variation in self-absorption effect is significant with various angular distributions. It is difficult to compare the simulated self-absorption effect with the experimentally used radioisotope source because Guo et al. did not report the source dimensions. They reported an electroplated Ni-63 was used in the experiment. It was assumed in this analysis that the source has a surface area of $4 \times 4 \text{ mm}^2$ similar to the p-n junction silicon semiconductor. The source thickness was estimated by MCNP simulations using the problem setup shown in Fig. 16 to justify the self-absorption effect in this analysis. A Ni-63 point source was moved from the surface along the thickness direction at different source positions of 0.01 μm, 0.05 μm, 0.1 μm, and up to 1.5 μm with an increment of 0.1 μm. The energy absorbed in the source was estimated using full beta energy spectrum for different source positions along with different angular distributions of 0° and 90°. Fig. 17 shows the percentage of beta energy absorbed along with uncertainty bars in the Nickel source for different source positions and angular distributions. It indicates that about 50% of the beta particle energy was absorbed within 1–1.5 μm thickness of Ni-63 due to the very high nickel density of 8.91 g/cm^3 . The estimated thicknesses for 0.25 mCi and 1 mCi activities of Ni-63 are 0.03 μm and 0.12 μm, respectively, based on the specific activity of 56.18 Ci/g for Ni-63. Fig. 17 shows that about 15% and 28.5% of the beta particle energy is absorbed for approximately 0.1 μm and 0.4 μm thicknesses, respectively, for 90° angular variation, which is more realistic. These percentages of energy absorption in the source are comparable to the percentages of reduction in battery outputs for self-absorption effect considered in Table 3 and Table 4. However, the thicknesses are higher than the theoretical thicknesses estimated from the specific activity. The isotope dilution factor, which is very high for a Ni-63 source (Gui et al., 2016), could be the reason for this. Assuming about 30% of Ni-63 by atoms in the source

Table 3
Results for full beta particle energy spectrum MCNP input with 0° angular distribution.

Activity	0.25 mCi			1 mCi		
	Simulation results (case 1)		Experimental results (case 1)	Simulation results (case 2)		Experimental results (case 2)
	No self-absorption	59% self-absorption		No self-absorption	65.1% self-absorption	
Short circuit current (nA)	1.73 ± 0.0073	0.71 ± 0.003	0.71	6.91 ± 0.0291	2.41 ± 0.0102	2.41
Open circuit voltage (mV)	152 ± 0.1	129.1 ± 0.1	64	187.8 ± 0.1	160.6 ± 0.1	115

Table 4
Results for full beta energy spectrum MCNP input with 90° angular distribution.

Activity	0.25 mCi			1 mCi		
	Simulation results (case 1)		Experimental results (case 1)	Simulation results (case 2)		Experimental results (case 2)
	No self-absorption	15% self-absorption		No self-absorption	28.5% self-absorption	
Short circuit current (nA)	0.84 ± 0.0053	0.71 ± 0.0045	0.71	3.36 ± 0.0211	2.41 ± 0.0151	2.41
Open circuit voltage (mV)	133.5 ± 0.2	129.3 ± 0.2	64	169.2 ± 0.2	160.6 ± 0.2	115

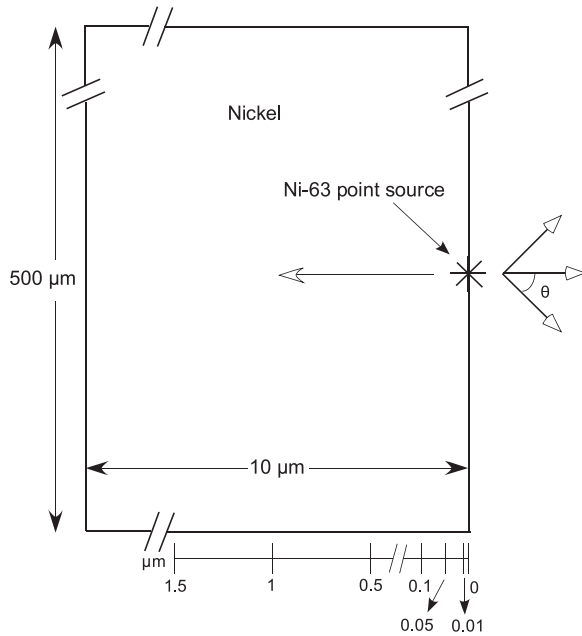


Fig. 16. Schematic diagram of Ni-63 source in MCNP simulation.

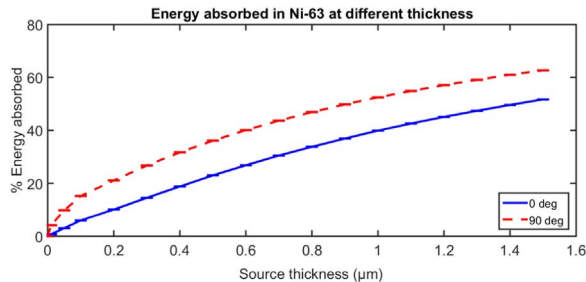


Fig. 17. Energy absorbed within the source (Ni-63) due to self-absorption.

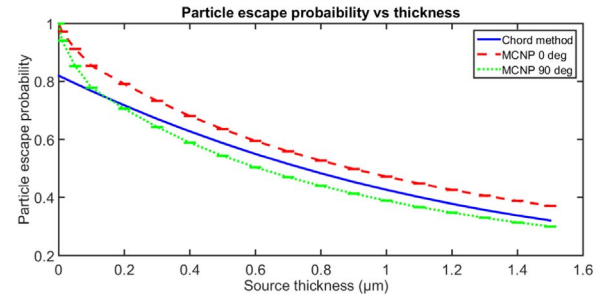


Fig. 18. Comparisons of particle escape probability for Ni-63 source.

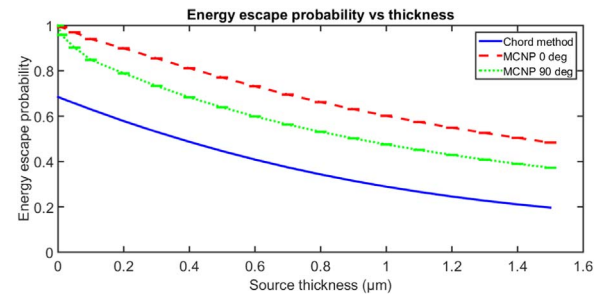


Fig. 19. Comparisons of energy escape probability for Ni-63 source.

would explain the estimates of higher thicknesses. The primary effect of self-absorption is to reduce the number of beta particles reaching the semiconductor surface and to let mainly higher energy particles get through. The net effect is to reduce the total energy available for deposition in the semiconductor which directly affects the betavoltaic battery output. This shows that the self-absorption of the beta particles coming from the source can have a significant impact on betavoltaic battery outputs. The effects of self-absorption and variation in beta particle angular distribution need to be further investigated and justified for betavoltaic battery model development. A self-absorption model was developed by Gui et al. (2016) using the direct chord method for a Ni-63 source. The model predicted the particle escape probability and the energy escape probability with thickness for Ni-63. The particle escape probability and the energy escape probability for Ni-63 from the chord method were compared with the MCNP results for both 0° and 90° distributions in Figs. 18 and 19 respectively. Both Figs. 18 and 19 show that the particle and energy escape probability are

lower for 90° distribution due to more inelastic and elastic scattering as well as higher absorption in the source. It is noted that the particle and energy escape probabilities are approximately 1 for a very thin source as most of the particles can escape. However, the chord model predicted lower particle and energy escape probabilities for a very thin source. Fig. 18 shows that the particle escape probability predicted by the chord model fell in between the MCNP results of 0° and 90° distributions. However, the energy escape probability predicted by the chord model was much lower than that of the MCNP results as shown in Fig. 19. The deviation in results could be caused by ignoring the spectrum modification and the assumption of an infinite slab in the chord model.

Besides the self-absorption effect, there could be other reasons for the deviation between the simulated and experimental results. A semiconductor device model with simplified assumptions such as uniform doping, negligible electric field in p-type and n-type regions, no recombination in the depletion region, full ionization of dopants, and a Boltzmann approximation was used. The quality of the device used in the experiment was not verified by measurements such as dopant concentrations and surface recombination velocity. Some important parameters in the model such as semiconductor junction depth, radioisotope source dimension, and device temperature were assumed, as they were not reported in the experiment. This will need to be further analyzed using other experimental data. It is noted that some experimental results on Ni-63 with silicon are available in the literature (Yakimov, 2016; Liu et al., 2012; Gao et al., 2013; Krasnov et al., 2015). However, it is difficult to compare the simulated results with the experimental results as the detail of the semiconductor parameters including the source dimension were not reported. Therefore,

measurements of all parameters in the experiment will be useful to validate, improve, and optimize the model.

7. Conclusions

Beta particle penetration depths using MCNP with different types of input energy were estimated for Ni-63 in silicon. The penetration depths from MCNP for inputs of monoenergetic beta particle average energy and monoenergetic beta particle maximum energy were validated by the Katz-Penfold range equation. However, using the actual beta energy spectrum produces significantly different beta particle penetration depths. Factoring in the isotropic angular variation of the beta particle emission reduces the penetration depth and increases the backscattering coefficient up to 0.46. Thus, the key point is when designing a betavoltaic battery a more realistic beta particle transport model needs to be used in order to choose an optimum junction depth for maximum collection of EHPs generated in the semiconductor in order to maximize battery outputs. The exponential distribution of EHP generation from MCNP results were used to solve the drift-diffusion equation of the semiconductor model. The predicted results such as the short circuit current density, the open circuit voltage density, and the leakage current density varied depending on the junction depth of the semiconductor and the angular distribution of beta particles. The power output and device efficiency variations were significant with the junction depth. A shallow junction depth depending on the surface recombination velocity will increase the power output and efficiency. The results were overpredicted when no source self-absorption effect was considered. For isotopic emission of beta particles, a 15% self-absorption effect was observed for an activity of 0.25 mCi with a junction depth of 2 μm . However, the self-absorption effect further increased with higher activity as expected since the source thickness would be greater. Therefore, the energy deposition and penetration depth of beta particles in the semiconductor, the choice of junction depth, the angular variation in the emission of the beta particles, the backscattering of beta particles, and the self-absorption effect of the source need to be considered carefully in betavoltaic battery design.

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